

Sequence Diagrams

Control, Conditions, Iteration & Object Lifecycle

Focus of Control

Focus of Control (Activation Box)

The focus of control indicates the period during which an object is active and consuming resources.

Key Points

- Shows when an object is executing an operation
- Indicates sending or receiving messages
- Represented by a thin vertical rectangle on a lifeline

Exam Tip

Focus of control shows **when** an object is active, not **how** it works internally.

Axes and Flow in Sequence Diagrams

Diagram Axes

- **X-axis:** Objects and actors
- **Y-axis:** Time (top to bottom)

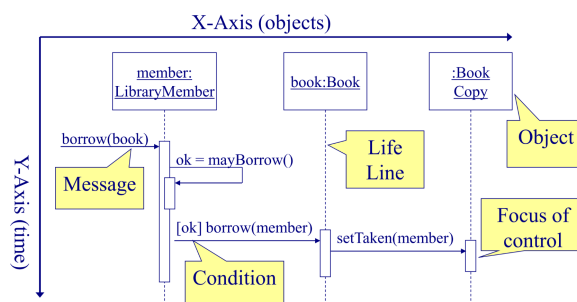


Figure 1: Sequence Diagram

Additional Concepts

- Object Life
- Messages
- Conditions
- Focus of Control

Control Information

Conditions

Condition

A condition controls whether a message is sent or not.

Syntax

[condition] messageName()

Example

[ok] borrow(member)

Meaning

The message is sent **only if the condition evaluates to true.**

Control Information: Alternatives

Conditional Message Flow

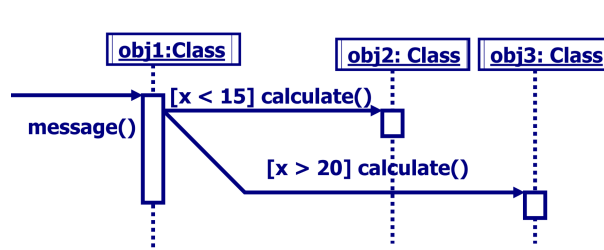


Figure 2: Control Information

Note

Sequence diagrams support only **simple alternatives**. For complex logic, use multiple diagrams.

Concurrency

Concurrency

Concurrency occurs when multiple objects execute independently or simultaneously.

Concept

- Messages can be processed in parallel
- Objects do not block each other

Example

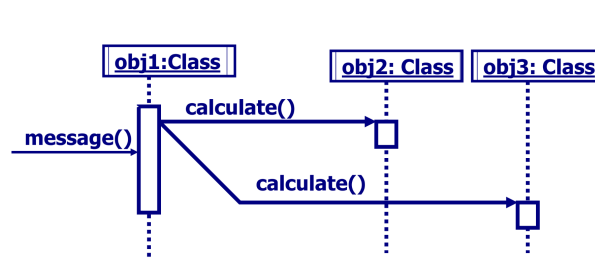


Figure 3: Concurrency

Iteration

Iteration

Iteration represents repeated message execution.

Syntax

*[condition] messageName()

Key Forms

- **Until condition** – exit when condition becomes true
- **While / For expression** – fixed or conditional repetition

Example

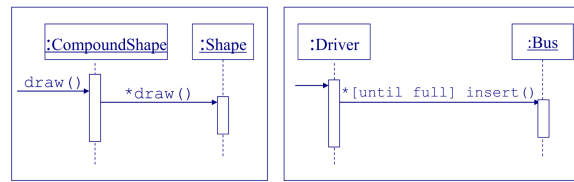


Figure 4: Iteration

Meaning

The message may be sent multiple times to one or more objects.

Object Creation

Object Creation

Objects can be created dynamically during a sequence using the «create» stereotype.

Rules

- The «create» message is sent to the object body
- A lifeline starts after creation
- The object can then send and receive messages

Preferred Notation

- :A → «create» :B

Object Destruction

Object Destruction

An object may be destroyed using the «destroy» stereotype.

Key Points

- An object may destroy another object or itself
- Destruction is shown by X at the end of the lifeline
- Avoid modeling destruction unless memory management is critical

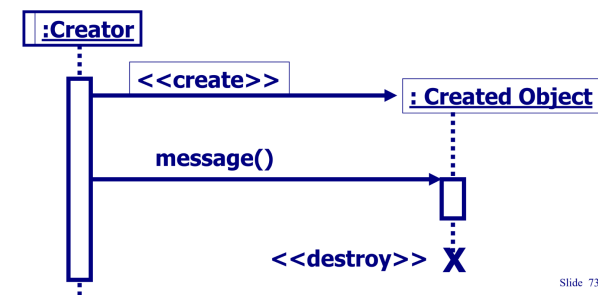


Figure 5: Object Creation and Destruction

Comprehensive Example: Traffic Violation Lookup

Scenario Description

- Clerk initiates lookup
- ID is retrieved
- Database is queried
- Violation object is created and returned
- Result is displayed

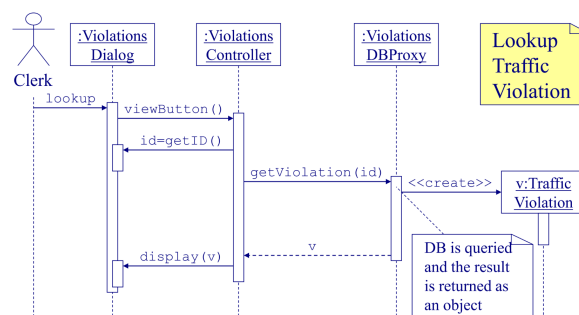


Figure 6: Traffic Violation Lookup

Key Concepts Demonstrated

- Message passing
- Object creation using «create»
- Data returned as objects

Design Advice

Do not use sequence diagrams for detailed algorithms. Use activity diagrams, pseudo-code, or state charts instead.